Thermal Conduction

Exam code: 0625 & 0972

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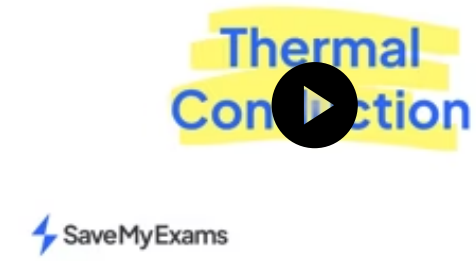
Updated on 28 September 2024



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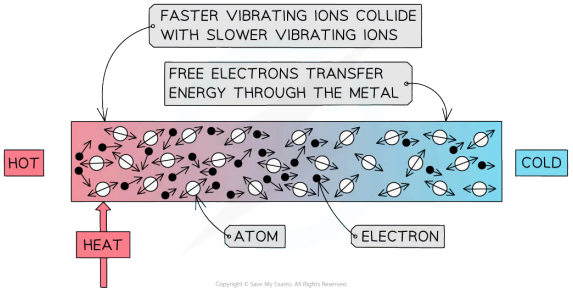
Thermal conduction in solids

Extended tier only

- Conduction is the transfer of heat from one region to another through particle vibrations and the movement of free electrons
- Conduction is the main method of thermal energy transfer in **solids**
- Metals are the best thermal **conductors**
  - This is because they have a high number of **free electrons**



Conduction: the atoms in a solid vibrate and collide, transferring energy

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- When a substance is heated, the atoms, or ions, **start to move around (vibrate) more**
  - The atoms at the hotter end of the solid will vibrate more than the atoms at the cooler end
  - As they do so they **bump into each other**, transferring energy from atom to atom
  - These collisions transfer internal energy until **thermal equilibrium** is achieved throughout the substance
  - This occurs in **all solids**, metals and non-metals alike
- Additionally, if a solid contains free electrons, these can travel through the material freely and transfer thermal energy through collisions

Thermal conduction in liquids & gases

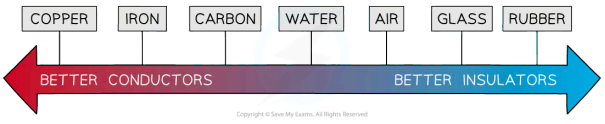
Extended tier only

- For thermal conduction to occur the particles need to be close together so that when they vibrate the vibrations are passed along
- This does not happen easily in fluids
  - In liquids particles are close, but slide past each other
  - In gases particles are widely spread apart and will not 'nudge' each other when they vibrate
- Both types of **fluid**, liquids and gases, are poor conductors of heat

Relative thermal conductivity

Extended tier only

- Conductors tend to be **metals**
  - Better thermal conductors are those with delocalised electrons which can easily transfer energy
  - This means that there is a **wide range** of thermal conductivity



Different materials have different levels of conductivity

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